Two-link robot arm: PID simulation

ME3202 Lab 7 study | Gajendra, 2301ME18

Abstract - short summary

This report presents an independent two-degree-of-freedom planar arm simulation inspired by the course manipulator lab. A gravity-compensated PID controller tracks two step joint targets. Link dimensions and the resulting angles are illustrative simulation outputs, not physical robot measurements.

Purpose and background

The course lab studies M(q) qdd + C(q, qd) qd + G(q) = torque and PID tracking. The portfolio code implements that structure in Python for a reproducible step-response study.

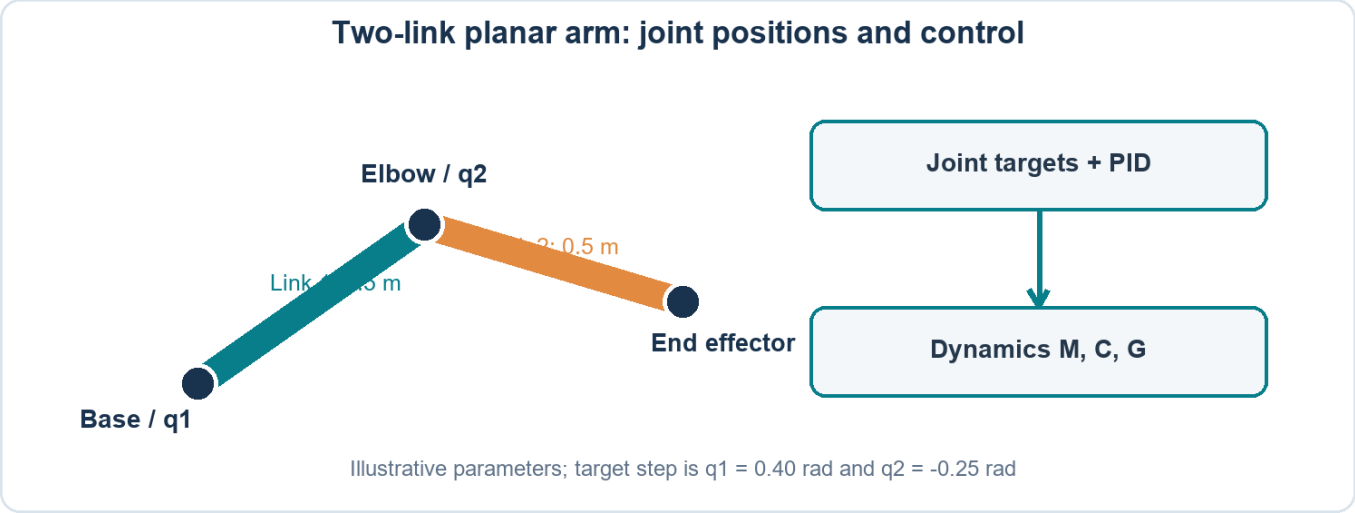


Figure 1. Two-link geometry and controller relationship. Link lengths are model parameters.

Methods - how the study was carried out

- Set link lengths to 0.5 m and masses to 1.0 kg each; form symmetric inertia, velocity coupling and gravity terms.
- Apply joint PID with gravity compensation, integral clamp and +/-15 N m torque limits.
- Integrate dynamics at 0.002 s steps for 3.0 s; compare q1 and q2 with 0.40 and -0.25 rad targets.

Two-link robot arm: PID simulation | Work and results

What was done

The items below describe the work represented by this repository and the course context.

• Gajendra reports completing the corresponding manipulator lab at IIT Patna.

• The portfolio adds a separate Python simulation, CSV response and tests of inertia and tracking.

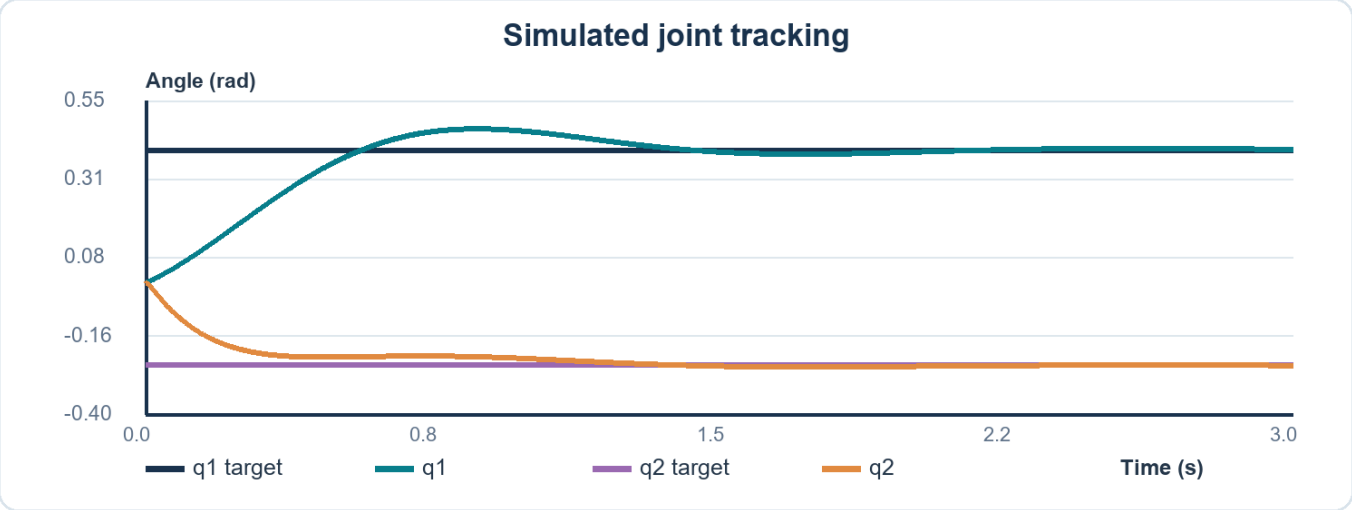


Figure 2. Joint trajectories generated by the Python model with default gains.

Result and interpretation

At 3.0 s, q1 = 0.4034 rad and q2 = -0.2514 rad. Absolute step errors are 0.0034 and 0.0014 rad. The response illustrates model-based tracking under the chosen parameters.

Limits and next validation step

This is a step-reference study only; the original lab also included ramp and sinusoidal inputs. Joint friction, backlash, encoder quantization and payload changes are not validated. Hardware work would require parameter identification, torque calibration and safe travel limits.

Reproduce it

Run: python projects/me3202_labs/two_link_pid/simulate.py

Open project code on GitHub